along $\hat{m}$. Prepare a spin so that the apparatus reads +1. Then, without disturbing the spin, rotate the apparatus to the direction $\hat{n}$, as shown in Fig. 1.4. A new experiment on the same spin will give random results ±1, but with an average value equal to the cosine of the angle between $\hat{n}$ and $\hat{m}$. In other words, the average will be $\hat{n} \cdot \hat{m}$.

The quantum mechanical notation for the statistical average of a quantity Q is Dirac's bracket notation $\langle Q \rangle$. We may summarize the results of our experimental investigation as follows: If we begin with $\mathcal{A}$ oriented along $\hat{m}$ and confirm that $\sigma = +1$, then subsequent measurement with $\mathcal{A}$ oriented along $\hat{n}$ gives the statistical result

$$\langle \sigma \rangle = \hat{n} \cdot \hat{m}.$$

What we are learning is that quantum mechanical systems are not deterministic—the results of experiments can be statistically random—but if we repeat an experiment many times, average quantities can follow the expectations of classical physics, at least up to a point.

1.4 EXPERIMENTS ARE NEVER GENTLE

Every experiment involves an outside system—an apparatus—that must interact with the system in order to record a result. In that sense, every experiment is invasive. This is true in both classical and quantum physics, but only quantum physics makes a big deal out of it. Why is that so? Classically, an ideal measuring apparatus has a vanishingly small effect on the system it is measuring. Classical experiments can be arbitrarily gentle and still accurately and reproducibly record the results of the experiment. For example, the direction of an arrow can be determined by reflecting light off the arrow and focusing it to form an image. While it is true that the light must have a small enough wavelength to form an image, there is nothing in classical physics that prevents the image from being made with arbitrarily weak light. In other words, the light can have an arbitrarily small energy content.